use case description:

1-
Use Case Name: Track Project Views
Primary Actor: Interior Designer

Description: Allows an interior designer to see how many times their project has been viewed by prospective customers.

Precondition: The designer is authenticated, has permission to view the project analytics, and the system has a functioning interaction-tracking mechanism.

Trigger: The designer selects a project to view its statistics.

Basic Flow: The designer selects a project. The system verifies access, retrieves valid view events, calculates the total views, and displays the result.

Alternative Flows: If the project has no views, display zero. If a date range is selected, count only views within that period.

Exception Flows: If access is denied or the tracking data cannot be retrieved, display an appropriate error instead of the count.

2-
Use Case Name: Track Unique Project Viewers
Primary Actor: Interior Designer

Description: Allows the designer to identify the number of distinct visitors who have viewed their project, rather than simply counting every visit.

Precondition: The designer is authorised and the system records suitable visitor identifiers for eligible views.

Trigger: The designer requests unique-viewer statistics.

Basic Flow: The system retrieves valid project view events, groups them by visitor identifier, counts the distinct visitors, and displays the result.

Alternative Flows: Repeated views from the same identified visitor count only once during the selected period. If no eligible viewers exist, display zero.

Exception Flows: If the system cannot reliably identify unique visitors or retrieve the data, indicate the limitation or display an error.

3-
Use Case Name: Track Project Shortlists
Primary Actor: Interior Designer

Description: Allows a designer to see how many distinct customers have added their project to a shortlist or favourites during the reporting period.

Precondition: The designer has access to the project analytics and shortlist interactions are being recorded.

Trigger: The designer requests shortlist statistics.

Basic Flow: The system verifies project access, retrieves valid shortlist-add events, counts distinct customers who added the project, and displays the count.

Alternative Flows: Repeated shortlist-add events from the same customer are counted once within the period. No qualifying events result in zero.

Exception Flows: If a required customer identifier is missing or event retrieval fails, the system must not present an unreliable count.

4-
Use Case Name: Track Project Shortlist Rate
Primary Actor: Interior Designer

Description: Calculates the proportion of unique project viewers who added the project to their shortlist.

Precondition: The designer is authorised and valid view and shortlist events can be correlated.

Trigger: The designer requests the shortlist rate.

Basic Flow: The system retrieves valid view and shortlist interactions, determines the number of distinct viewers and how many of those viewers shortlisted the project, calculates the percentage, and displays it.

Alternative Flows: If the project has no eligible viewers, display N/A rather than calculating a percentage. Repeated actions from the same viewer are counted only once.

Exception Flows: If visitor identities cannot be correlated reliably or the underlying data is unavailable, the system displays an error or unavailable status.

5-
Use Case Name: Track Customer Enquiries

Primary Actor: Interior Designer

Description: Allows the designer to identify the number of customer enquiries associated with each project.

Precondition: Enquiry interactions are tracked and linked to projects, and the designer has permission to view the analytics.

Trigger: The designer requests project enquiry statistics.

Basic Flow: The system verifies access, retrieves accepted enquiry events associated with the selected project, counts the qualifying enquiries, and displays the total.

Alternative Flows: Display zero when the project has no enquiries. Apply the selected date range when provided.

Exception Flows: Exclude enquiries that cannot be attributed to the project. Display an error if the required information cannot be retrieved.

6-
Use Case Name: Track Project Engagement Time
Primary Actor: Interior Designer

Description: Allows the designer to view the average active engagement time customers spend exploring their projects.

Precondition: The system records engagement durations and the designer is authorised to access the project.

Trigger: The designer requests engagement-time statistics.

Basic Flow: The system retrieves valid engagement records, excludes invalid durations, calculates the average active engagement time, and displays the result.

Alternative Flows: If no measurable sessions exist, display N/A. Multiple valid sessions are included in the calculation.

Exception Flows: Invalid or missing duration values are excluded according to the measurement rules. If the tracking data is unavailable, display an error.

7-
Use Case Name: View Project Analytics Dashboard
Primary Actor: Interior Designer

Description: Provides a centralised dashboard displaying interest metrics for the designer's projects.

Precondition: The designer is authenticated and authorised to access the requested analytics.

Trigger: The designer opens the analytics dashboard.

Basic Flow: The system verifies the designer's access rights, retrieves valid interaction records, calculates the relevant metrics from US01–US06, and displays the results for accessible projects.

Alternative Flows: If the designer has no projects, display an empty dashboard. If there are no interactions, display zero or N/A where appropriate. A selected date range refreshes the metrics.

Exception Flows: If the designer is unauthorised or data retrieval fails, display an appropriate error without exposing restricted information.

8-
Use Case Name: Analyse Interest Trends Over Time
Primary Actor: Interior Designer

Description: Allows the designer to visualise how project views, shortlists, and enquiries change over time.

Precondition: The designer is authorised and the system has valid timestamped interaction records.

Trigger: The designer selects the interest-trends view.

Basic Flow: The system verifies project access, retrieves accepted interactions for the selected period, groups events into reporting intervals, and displays the resulting graphs.

Alternative Flows: The designer may change the reporting period or time grouping. Periods without activity are displayed as zero where applicable.

Exception Flows: An invalid date range is rejected. If the required data cannot be retrieved, the system displays an error rather than an unreliable graph.

9-
Use Case Name: Compare Project Performance
Primary Actor: Interior Designer

Description: Allows the designer to compare views, shortlist counts, and enquiry counts across their different projects.

Precondition: The designer is authorised to access at least two projects.

Trigger: The designer selects two or more projects for comparison.

Basic Flow: The system verifies access to each selected project, retrieves valid interaction records, calculates comparable metrics for a common reporting period, and displays the results side by side.

Alternative Flows: Additional authorised projects can be included. A project with no activity shows zero for applicable metrics.

Exception Flows: If fewer than two projects are selected, the system prompts for another selection. If an unauthorised project is included or analytics retrieval fails, the comparison is not displayed as valid.

10-
Use Case Name: Filter Analytics by Date Range
Primary Actor: Interior Designer

Description: Allows a designer to restrict analytics results to a specific period, such as the past week, month, or year.

Precondition: The designer is viewing an analytics feature they are authorised to access.

Trigger: The designer selects a date range or period preset.

Basic Flow: The designer chooses a reporting period. The system validates the range, retrieves the relevant accepted interactions, recalculates the affected metrics, and refreshes the analytics display.

Alternative Flows: A predefined period is converted into start and end dates. If there are no interactions in that period, the system displays zero or N/A as appropriate.

Exception Flows: Invalid date ranges are rejected. If access is denied or data retrieval fails, the system does not display restricted or misleading results.

11-
Use Case Name: Exclude Automated Traffic and Duplicate Events

Primary Actor: Interior Designer (beneficiary); Customer-facing Website Module (initiating system actor)

Description: Ensures that known automated traffic and duplicate interaction events are excluded from the analytics, improving the reliability of the project's interest metrics.

Precondition: Website modules generate interaction events containing the information needed for validation and duplicate detection.

Trigger: A website module submits a project interaction event.

Basic Flow:
1. The website module sends a tracking event containing project information and an event identifier or fingerprint.
2. The system validates the required event information.
3. The system checks whether the event originates from known automated traffic.
4. The system checks whether the event has already been accepted.
5. If the event passes both checks, the system records it.
6. The recorded interaction becomes available for analytics calculations.
Alternative Flows:

A1 — Automated Traffic Detected: The system identifies the event as originating from known automated traffic and excludes it from analytics.

A2 — Duplicate Event Detected: The system finds that an identical tracking event has already been recorded and skips the duplicate.

Exception Flows:

E1 — Invalid Event Information: The event lacks required fields and is rejected or quarantined.

E2 — Validation or Storage Failure: The system records the failure and does not assume the event was successfully accepted.

12-
Use Case Name: Authorise Project Analytics Access

Primary Actor: Interior Designer

Description: Restricts project analytics so that only authenticated users with the necessary permissions can access the relevant business performance data.

Precondition: User identity information and project ownership or permission records are available.

Trigger: A user requests analytics information for one or more projects.

Basic Flow:
1. The interior designer requests project analytics.
2. The system verifies that the user is authenticated.
3. The system identifies the projects being requested.
4. The system verifies that the designer is authorised to access the analytics for every requested project.
5. If access is authorised, the system retrieves the analytics.
6. The system displays the requested information.

Alternative Flows:

A1 — Multiple Projects Requested: The system checks access rights for every selected project before returning comparison or dashboard results.

A2 — Authorised Access: The requested dashboard, trend, comparison, or individual metric operation continues normally.

Exception Flows:

E1 — Unauthenticated User: The system rejects the request and requires the user to sign in.

E2 — Unauthorised User: The system refuses access to the requested analytics without revealing the restricted data.

E3 — Authorisation Service Unavailable: The system does not return analytics until it can verify permission.